

The Short Mood and Feelings Questionnaire From Adolescence to Emerging Adulthood: Measurement Invariance Across Time and Sex

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Adolescence to emerging adulthood is a critical period for the onset of depressive symptoms. Understanding symptom change during this period is thus of great clinical relevance. This understanding is, however, based on the premise of the accurate measurement of depressive symptoms across time and sex, typically untested in applied research. The present study investigated longitudinal and sex measurement invariance (MI) of the Short Mood and Feelings Questionnaire (SMFQ), a widely used unidimensional 13-item measure of self-reported depressive symptoms. We employed 10 waves of the Avon Longitudinal Study of Parents and Children, a population-based study in South-West England ($N = 7,364$; ages 11–26). The SMFQ exhibited increasing consistency with age: Scalar longitudinal MI was not supported by all indices in models that included ages 11 and 13, but strict MI was established from ages 14–26. At each wave, at least partial strict MI across sex was established. Sum score models with equal weightings had acceptable fit, and good reliability which was equivalent to reliability using differential weightings. External validity for sum scores was also comparable to factor scores. Thus, sum scores seem an appropriate, practical choice in many settings. Overall, findings support the use of SMFQ in assessing change in depressive symptoms from adolescence into emerging adulthood, specifically ages 14–26. Some caution is necessary when comparing the construct at ages 11–13 with ages greater than 17, when measurement models were not fully invariant. This research informs epidemiological and clinical studies on the applicability of the SMFQ across time and sex.

Public Significance Statement

Understanding depressive symptom change from early adolescence into emerging adulthood is important for intervention purposes. By investigating measurement invariance over age and sex for the Short Mood and Feelings Questionnaire, a widely used measure of self-reported depressive symptoms, this research informs epidemiological and clinical research on the robustness and consistency of depressive estimates obtained using this measure.

Keywords: ALSPAC, depression, adolescents, Short Mood and Feelings Questionnaire, longitudinal measurement invariance

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Functional impairment, global disease burden, and medical costs associated with depression necessitate accurate screening of this disorder (Vos et al., 2015). This is especially relevant for adolescence, a period characterized by a myriad of social, biological, and psychological changes and a rapid increase in the incidence of depression (Dahl et al., 2018; Thapar et al., 2022). The self-report Mood and Feelings Questionnaire (MFQ; Angold et al., 1995) has been recommended by the National Institute for Health and Clinical Excellence guidelines as a screening tool for childhood and adolescent depression (Hopkins et al., 2015; Middleton et al., 2005).

In epidemiological studies, the Short Mood and Feelings Questionnaire (SMFQ) is often used to assess depressive symptoms because it is easy to administer and SMFQ scores demonstrate good reliability and validity in clinical and nonclinical populations (Angold et al., 1995). Accordingly, SMFQ sum scores are widely used as an estimation of the distribution of depressive symptoms in different populations. However, it remains unknown whether the SMFQ measures the same latent construct across time and sex in key phases of development from early adolescence into emerging adulthood.

The SMFQ

The SMFQ was developed with core items extracted from the MFQ to provide a parsimonious unidimensional instrument based on diagnostic classifications (Angold et al., 1995). SMFQ items were selected based on their correlations with the MFQ total scores, factor loadings, and their correlations with clinical status and depression diagnosis (Angold et al., 1995). The SMFQ consists of 13-items assessing past 2-week depressive symptoms. Most SMFQ items tap into cognitive (i.e., negative self-evaluation) and affective components of depressive symptoms, but also pertain to tiredness, restlessness, and poor concentration. The SMFQ has been found to be unidimensional in separate samples from ages 6 to 18 years in girls and boys (Thabrew et al., 2018; Turner et al., 2014).

Many studies have tested the psychometric properties of the SMFQ from ages 6 to 25. Both construct and criterion validity of the SMFQ scores have been established in different samples aged 7–18 (Sharp et al., 2006; Thapar & McGuffin, 1998; Turner et al., 2014). In cross-sectional analyses, SMFQ scores displayed good reliability (Cronbach's α : .80–.91) and validity in cohorts aged 8–16 (Angold et al., 1995), 11–13 (Rhew et al., 2010), 18 years (Turner et al., 2014), and at Age 25 (Eyre et al., 2021). Although SMFQ items are not fully reflective of diagnostic criteria, SMFQ scores demonstrated acceptable to good sensitivity (.60–.86), specificity (.61–.87), and discrimination ability (area under the curve: .72–.84) in children and adolescents ages 6–17 years when tested against clinical interviews (Angold et al., 1995; Rhew et al., 2010; Thabrew et al., 2018; Thapar & McGuffin, 1998). Symptoms not covered in diagnostic classifications such as self-hate and not being loved were particularly predictive of subsequent depressive symptom development (McKenzie et al., 2011).

In sum, SMFQ offers a psychometrically sound and parsimonious tool that is often utilized in child and adolescent mental health research: SMFQ is widely used in epidemiological studies until the age of 18, and in some instances until the age of 26 (Kingsbury et al., 2016; Kwong, 2019). However, it cannot be assumed that SMFQ measures the same construct across these ages and by sex, given differences in the expression of depression across development and

sex (Thapar et al., 2022). As SMFQ estimates inform epidemiology, measurement invariance (MI) across age and sex must be established to ensure consistency of depressive estimates obtained using this measure.

MI During Development

Changes in psychobiological functioning from early adolescence to emerging adulthood are relevant to depression and its expression (Dahl et al., 2018; Thapar et al., 2022). Testing consistent measurement of the depression construct is particularly relevant for mood self-reports because these symptoms are often not directly observable. When MI is violated, differences in measurement components such as factor loadings, item thresholds, or error terms can lead to biased test scores and erroneous conclusions (Liu et al., 2017). MI could be violated if the depression construct has a different structure across age or sex. For instance, if SMFQ displayed a two-factor structure (e.g., reflecting somatic and cognitive symptoms) at earlier ages while becoming unidimensional at later stages, this would suggest that somatic symptoms are more reflective of depression at older ages. MI could also be violated if single symptoms show differential functioning across age or sex, for example, exhibiting floor or ceiling effects in one of the sexes, being prone to greater recall bias at younger ages, or changing in their underlying meaning across development.

To establish MI across time or sex, one first investigates whether the factor structure is equivalent across time or sex (configural MI). Next, factor loadings are additionally constrained to be equal across time or sex to test whether the symptoms relate to the latent construct of depression in the same way over time (weak/metric MI). Then, item thresholds are additionally constrained to be equivalent to examine whether the observed thresholds conditional on the latent factor do not differ across time or sex (strong/scalar MI). Last, residual variances of the items are additionally constrained to be equal to gauge whether the amount of variance in the items not accounted for by the latent depression factor is the same across time or sex (strict/unique factor MI). Only if these conditions are satisfied is it truly possible to establish that observed mean-level differences across time or sex reflect true differences in the depression construct (Liu et al., 2017). Thus, epidemiological research can only inform clinical practice when any changes in the means, variances, or covariances of the observed scores are assessed without bias across time.

Potential Sources of Noninvariance Across Adolescence and Emerging Adulthood

Depression symptoms can change in phase with developmental changes across domains, such as cognition, hormones, and relationships (Dahl et al., 2018). For instance, self-concept rapidly evolves over adolescence (Coughlin & Robins, 2017). Thus, the mental representation of items such as *being a bad person* may be reflective of depression to a different degree across development and thus have different factor loadings over time. Furthermore, appetite/weight change, energy loss, and insomnia are common depressive symptoms in adolescents, whereas in adulthood, depression has been more commonly characterized by feelings of depression (e.g., “feeling miserable”) and anhedonia (e.g., “not enjoying anything”; Rice et al., 2019). This could be reflected in lower factor loadings on the latter items at younger ages, resulting in these items being poor

discriminators of the latent construct among younger people. In addition, this may be reflected in higher thresholds of these items (e.g., fewer endorsements) in younger ages. Moreover, items assessing feelings of tiredness, restlessness, or tearfulness may be linked to pubertal development and may thus be noninvariant over time (Lewis et al., 2018). Older adolescents may be more affected by tiredness, and so more readily endorse this item, resulting in a lower item threshold. In sum, we need to establish whether SMFQ items are equally reflective of depression during a life phase that is characterized by ongoing developmental change, across identity, social, and biological domains (Dahl et al., 2018).

Only one study could be identified that tested structural invariance of the SMFQ across childhood and adolescence, but this was limited to boys aged 8–16 (Messer et al., 1995). This study identified the same unidimensional factor structure across age groups (configural MI) and high factor loading correlations over time (indicating loading stability without formally testing metric MI). However, mean-level factor loadings increased with age indicating more item error variance in younger ages.

Potential Sources of Noninvariance Across Sex/Gender

MI across development is also confounded by changes in depressive symptoms across sex or self-reported gender which emerge during the ages of 12–15, when females display an earlier and steeper increase in depressive symptoms compared to males (Patalay & Fitzsimons, 2018; Shore et al., 2018; Solmi et al., 2021). Overall, women report higher rates of major depression compared with men, with a ratio around 2:1 (for a meta-analysis see Salk et al., 2017). These differences have been attributed to a complex combination of social, biological, and psychological factors, such as pressure to satisfy gender roles (Anyan & Hjemdal, 2018), pubertal development (Lewis et al., 2018), and differences in rumination tendencies (Nolen-Hoeksema & Aldao, 2011), respectively.

To date, it remains unclear whether the SMFQ measures the same underlying construct in females and males in these critical developmental phases. One study tested MI across gender in 17-year-old Norwegians and found that with the same underlying level of depression, girls had a lower threshold for reporting that they felt miserable, tired, and that they cried a lot, compared with boys (Lundervold et al., 2013). Boys had a slightly lower threshold to report that no one loved them. In another study of 8–12-year-olds from Spain, feeling tired had a higher probability of being endorsed by girls, whereas “nobody loved me” had a higher probability of being endorsed by boys (Espada et al., 2022). If some items are only related to depression in females but not males, and females endorse these items more often, this would inflate the composite scale scores for females only. Concluding that females have higher levels of depressive symptoms would thus not be justified because these items were initially not related to the construct of depression in males. In fact, such items may purely be an artifact of social norms in the way in which females and males express their distress instead of being equally reflective of depression across sex (Anyan & Hjemdal, 2018; Santor et al., 1994).

From early adolescence to emerging adulthood individuals grow into social roles and explore and adapt behavior in various contexts (Dahl et al., 2018). These experiences along with environmental expectations may shape the way in which females and males express their depressive feelings or thoughts across development. In addition, females tend to attend more to their emotions than boys (Neumann

et al., 2010), and items such as *feeling restless* or *having trouble concentrating* may be more readily expressed in the context of depressive symptoms in females compared to males. Testing whether these items reflect the depression construct equally across sex in different developmental stages is thus warranted.

Use of Sum Scores and External Validators

Many applied studies use SMFQ composite scores (Riglin et al., 2021), implicitly assuming a homogenous construct with symptoms contributing uniformly to the manifestation of depression (i.e., items having the same factor loadings or weightings). However, these assumptions need testing (McNeish & Wolf, 2020; Widaman & Revelle, 2022). For sum score totals to be valid, a one-factor model must provide an adequate fit to the data and even then, the reliability of a differentially weighted sum score might be stronger than the reliability of the equal weighting sum score (Widaman & Revelle, 2022). For an overall psychometric evaluation, it is therefore important to test whether a one-factor model fits the data adequately (with or without all factor loadings constrained to be equal), and test the comparative reliability of the equal-weighted versus differential-weighted sum scores (Widaman & Revelle, 2022). Finally, for comparative validity, it is relevant to test whether associations with external validators differ when factor scores are used compared to sum scores. If the former scoring method indicates a medium effect size with an external construct, whereas the latter method a small effect size, this may indicate some bias in the sum scores based on equal loadings.

The Present Study

We aimed to systematically examine MI of the SMFQ from early adolescence into emerging adulthood, covering the ages of 11–26 years. To this end, we used data from the Avon Longitudinal Study of Parents and Children (ALSPAC), a population-based study in the South-West of England (Boyd et al., 2013). Our research aims were threefold. Our *first research aim* was to test temporal MI. Here, we expected that the one-factor solution would be confirmed across all timepoints since this factor solution is consistently reported in the literature (Thabrew et al., 2018; Turner et al., 2014). We tested exploratorily whether strict MI could be confirmed for this factor solution across waves. Our *second research aim* was to test MI across sex at each wave. Again, we expected the one-factor solution to hold across waves for females and males. However, we speculated that items such as feeling miserable, tiredness, or crying might reflect different levels of the underlying construct (threshold noninvariance) at some timepoints across sex (Lundervold et al., 2013). The *third research aim* was to test the presumptions underlying sum score models (McNeish & Wolf, 2020; Widaman & Revelle, 2022): Would the one-factor model provide adequate fit to the data, and would reliability and validity differ across factor scores and sum scores? We expected the sum score model to fit less well than the factor score model but determined whether it still yielded acceptable model fit. To test external associations of the scoring methods, we drew on available constructs in the ALSPAC study with an established relationship with depression: sex (Shore et al., 2018), psychotic symptoms (Stochl et al., 2015), anxiety symptoms (Cummings et al., 2014), general health symptoms (Garcia-Cebrian et al., 2006), parental mental health

reports (Finsaas et al., 2018), and life events (Korten et al., 2012). Given that most constructs were assessed at different timepoints, we expected small-to-moderate correlations with the SMFQ scores. We explored whether the derived conclusions concerning these associations differed based on the scoring method.

Method

Sample

Details of the ALSPAC study have been described previously (Boyd et al., 2013; Fraser et al., 2013; Northstone et al., 2019). Briefly, 14,541 pregnant mothers were recruited into the ALSPAC study; sample refreshments when their children were Age 7 resulted in a sample of 15,454 pregnancies with data available after the age of 7, of which 14,901 participants were alive at 1 year of age resulting in the final sample. Ethical approval was given by ALSPAC Law and Ethics Committee and local Research Ethics Committees. Informed consent for the use of data collected via questionnaires and clinics was obtained from participants following the recommendations of the ALSPAC Ethics and Law Committee at the time. Compared to representative data from the U.K., ALSPAC participants were more likely to have higher education, be white, and were less likely to receive free school meals (Boyd et al., 2013). SMFQ was assessed at ten occasions from ages 11 to 26 years (Supplemental Table S1 for demographics). Study data were collected and managed using Research Electronic Data Capture (REDCap) electronic data capture tools hosted at the University of Bristol (Harris et al., 2009). REDCap is a secure, web-based software platform designed to support data capture for research studies. The study website contains details of all the data that is available through a fully searchable data dictionary and variable search tool (<http://www.bristol.ac.uk/alspac/researchers/our-data>). Data and materials cannot be made openly available as data need to be paid for. Researchers can request the data at <http://www.bristol.ac.uk/alspac/researchers/access/>. The present study was not preregistered.

Measures

SMFQ

The 13-item SMFQ (Angold et al., 1995) was designed for children and adolescents to assess depressive symptoms over the last 2 weeks. Response categories are *not true* (0), *sometimes* (1) and *true* (2) leading to total scores ranging from 0 to 26. The SMFQ has been commonly used in epidemiological studies (Kingsbury et al., 2016) and its scores have demonstrated good reliability and validity (Turner et al., 2014). Based on testing against the Clinical Interview Schedule-Revised, a cut-off value above 11 has been considered indicative of depression (Thabrew et al., 2018). The SMFQ was completed using a computer at a clinic visit (Wave 1, 2, 3, and 5) or via pen-and-paper at home (Kwong, 2019).

External Validation Variables (Research Aim 3)

Sex

Participants' parents reported binary sex (male/female) at baseline only. Even if sex MI is established, mean levels are expected to differ by sex (Shore et al., 2018).

General Health

One question assessed the participants' self-reported general health. This question was assessed on 5-point Likert-type rating scale ranging from 1 (*excellent*) to 5 (*poor*) when participants were 18.7 years, 21 years, and 22 years.

Generalized Anxiety Disorder Scale

The Generalized Anxiety Disorder Scale is a well-validated self-report anxiety scale containing seven items (Löwe et al., 2008). A 4-point Likert-type rating scale was used ranging from 1 (*not at all*) to 4 (*nearly every day*). It was administered when participants were approximately 22 years of age (Wave 7 in the present study). Internal consistency in the present analyses was good, $\omega_t = .93$, $\alpha = .91$.

Psychotic Symptoms

The psychosis-like symptoms questionnaire (Horwood et al., 2008) assessed self-reported hallucinations and delusions via eight questions at Wave 4 (Age 17). Response options were (1) *no, never*, (2) *yes, maybe*, and (3) *yes, definitely*. Internal consistency in the present analyses was satisfactory, $\omega_t = .75$, $\alpha = .71$.

Strength and Difficulties Questionnaire (SDQ)

The Strength and Difficulties Questionnaire (SDQ) consists of 25 questions with five subscales (prosocial behavior, hyperactivity, emotional symptoms, conduct problems, peer problems, Goodman, 2001). SDQ was completed by the mothers for their children at Age 11 and 13. We used a total difficulties score based on the emotional, behavioral, attention and peer relationships subscales, and tested the emotional symptoms subscale separately as this scale should particularly be related to depression. Reliability and validity of the SDQ scores have been previously shown up to Age 17 (Goodman, 2001, $\alpha = .80-.82$).

Life Events Since 12 years

At 16 years, participants were asked about life events with a list with 22 events (*no/yes*). Events were, for example, *moving to a new neighborhood* or *death of a close friend* (see Kingsbury et al., 2016).

Analysis Strategy

Missingness

Analyses were conducted in R (R Core Team, 2021) using the *lavaan* package (Rosseel, 2012). The highest amount of missingness compared to Wave 1 numbers was 55% (3305/7364 complete) at Wave 7 (Supplemental Table S1). As previously reported, missingness was associated with being male, having mothers with poorer educational attainment, lower socioeconomic status, and having a younger mother (see Kwong, 2019). At all waves apart from Wave 3, higher SMFQ scores at Wave 1 were positively related to attrition, in line with findings that greater severity of depression predicts attrition and missingness (White et al., 2016). A wealth of further variables that were collected predicted missingness (Supplemental Table S2), bolstering support for data being missing at random (MAR; White et al., 2011). Given the ordinal MAR data, we used

the weighted least squares mean and variance adjusted (WLSMV) estimator for all models (Asparouhov & Muthén, 2010). Simulation studies of MI testing indicate that the WLSMV estimator (without auxiliary variables) produces relatively unbiased parameters and standard error estimates even with 50% MAR missingness when the sample size is 1,000 or more (Chen et al., 2020).

Tested Models

We tested a model including all 10 waves to examine whether MI holds for the entire period from 11 to 26 years. Based on theoretical considerations, we decided to test two different models if MI did not hold across this age range. This was an a priori decision to provide information for applied research to pinpoint whether there are specific periods across development where MI holds. First, we tested MI in adolescence (Waves 1–5), covering a distinct period from early adolescence (Age 11) until the age of 18 (mean age of 17.8), as many young people leave home when they are 18 (Dahl et al., 2018). Each of Waves 2 through 5 had responses by at least 61% ($N = 2,591$ complete cases) of Wave 1 participants (Table 1). Second, we conducted MI analyses separately in emerging adults (Waves 6–10), ages 19–26, as this reflects another distinct period of social and emotional development (Dahl et al., 2018), when many mental disorders continue to emerge (Solmi et al., 2021; 81% available data from Wave 10 with highest N compared to lowest N at Wave 7; $N = 1,240$ complete cases).

Factor Solution. Despite the previously reported one-factor solution of SMFQ from 6 to 18 years (Thabrew et al., 2018; Turner et al., 2014), we ran a series of exploratory factor analyses separately for each timepoint to ensure that we did not miss factor solutions at different ages. This procedure was data driven as the authors are not aware of other reported factor solutions for the SMFQ. The number of underlying factors was determined by the eigenvalues of the factors (i.e., parallel analyses, Horn, 1965, and inspecting the scree plot).

MI Testing (Research Aims 1–2). Temporal MI of the final factor solution was tested by comparing increasingly constrained models in a confirmatory factor analyses (CFA) framework in a single-group model with all waves in wide format for longitudinal MI (Liu et al., 2017; see Figure 1 for a schematic representation of the model). First, the factor structure was constrained to be invariant across time (configural MI). Second, as data were categorical, factor loadings, and item thresholds of the same items were constrained to be equal across waves (scalar MI). This was based on the premise that the probability of endorsing an item response category is jointly determined by the factor loadings and thresholds (Chen et al., 2020; Muthén & Muthén, 2017; Sass et al., 2014). Finally, item error variances were constrained to be equal over time to establish strict MI (unique factor MI). The residuals of the same symptoms were allowed to covary across waves. For model identification, we applied the steps outlined by Edossa et al. (2018) for categorical MI and theta parameterization was used. In all models, the comparative fit index (CFI) should be above .95 (.90 for acceptable fit) and the root-mean-square error of approximation (RMSEA) below .05 (below .08 for acceptable fit) to suggest good model fit (Hu & Bentler, 1999). We did not investigate differences of the χ^2 test statistic because they are likely significant given our large sample size (Liu et al., 2017). Next, we evaluated changes in these indices in the respective model compared to the previously highest established level of MI to establish the next level of MI (Edossa et al., 2018; Liu et al., 2017): $\Delta CFI \geq .010$

and $\Delta RMSEA \geq .007$ indicate substantial deterioration in model fit (Neufeld et al., 2022). When full MI was not supported, we tested for partial MI by iteratively relaxing constraints on parameters with the highest deviations (Liu et al., 2017). Next, we tested MI across sex (male vs. female). We performed multigroup modeling separately for each wave. The imposed constraints mirrored those of longitudinal MI testing, yet they were constrained to be equal across sex.

Sum Score Model Testing (Research Aim 3). We tested the appropriateness of sum score models by assessing model fit, reliability (internal consistency), and validity of the sum score versus factor models (Widaman & Revelle, 2022). *Model fit* was assessed when all factor loadings within each wave were set to be equal (McNeish & Wolf, 2020; Widaman & Revelle, 2022), and compared to the fit of the unconstrained configural model (McNeish & Wolf, 2020), using the same model comparison criteria as for MI testing above. Following Widaman and Revelle's recommendations (2022), *reliability* was estimated using omega total (ω_{total}), which does not assume equivalent factor loadings, as well as using Cronbach's α if the equal loadings model fit the data adequately. For assessment of *validity*, we extracted factor scores at each wave with the empirical Bayes method (Muthén, 1998–2004), and assessed time-varying associations with external validators compared to these associations with sum scores. To evaluate practical consequences of differences between the configural and most highly constrained MI models, we investigated to what extent the derived scores differed in their time-varying associations with external validators, depending on the level of longitudinal MI modeled. We derived two sets of factor scores: (a) factor scores from the longitudinal configural MI models (lowest restrictions) and (b) factor scores based on the highest level of longitudinal MI we established. Finally, we contemporaneously correlated sum scores with extracted factor scores to assess the consistency of what is used in common practice (sum scores) with the more accurate model (factor model; McNeish & Wolf, 2020). Given the large sample size, we set $\alpha = .001$ for all analyses.

Results

Descriptive Statistics

Descriptive statistics, internal consistencies, and scale composite scores over time are shown in Table 1. Across most waves, the item with the highest mean was *feeling miserable/unhappy*. The symptoms, *I hated myself*, *I was a bad person*, and *I did everything wrong* had a relatively low agreement over time, being significantly lower than most other items across waves. According to paired t tests, the symptoms *did not enjoy anything* and *feeling tired* showed significant mean increases across time ($p < .001$). The lowest SMFQ total mean score was found at Wave 2 (Age 13), the highest mean score at Wave 9 (Age 24). Skewness and kurtosis of most items were acceptable (< 3 for skewness and < 7 for kurtosis). Histograms indicate that all items had endorsement for all response options at all waves (Supplemental Figures S1–S10).

Factor Solution

Results suggested a clear one-factor solution based on scree plots and parallel analyses (Supplemental Table S3 and Figure S11). Therefore, we continued our MI analyses with a one-factor solution.

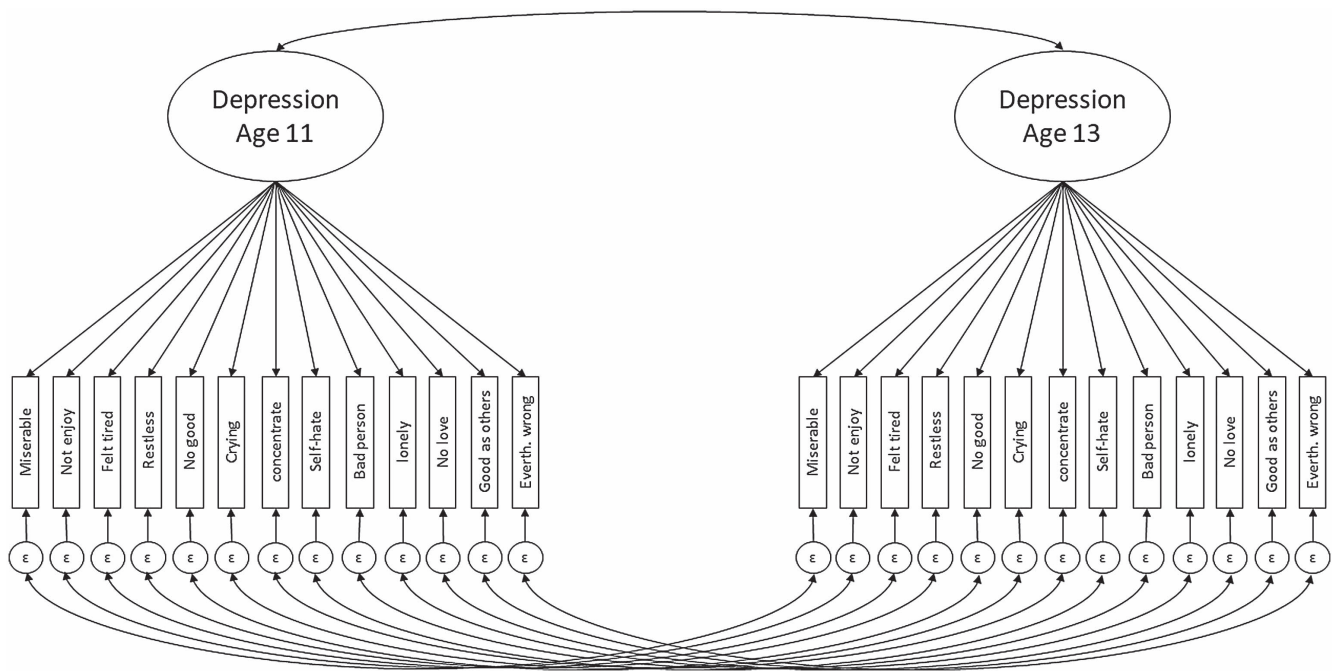
Table 1
Descriptive Statistics of the SMFQ Items, Sum Scores for Each Wave, and Internal Consistencies

<i>M (SD)</i>	Wave 1 (<i>N</i> = 7,364)	Wave 2 (<i>N</i> = 6,716)	Wave 3 (<i>N</i> = 6,019)	Wave 4 (<i>N</i> = 4,997)	Wave 5 (<i>N</i> = 4,497)	Wave 6 (<i>N</i> = 3,335)	Wave 7 (<i>N</i> = 3,305)	Wave 8 (<i>N</i> = 3,856)	Wave 9 (<i>N</i> = 3,915)	Wave 10 (<i>N</i> = 4,085)
1. Miserable/unhappy	0.60 (0.59)	0.69 (0.62)	0.71 (0.62)	0.93 (0.70)	0.84 (0.64)	0.98 (0.73)	0.79 (0.66)	0.82 (0.66)	0.90 (0.69)	0.84 (0.71)
2. Did not enjoy anything	0.15 (0.39)	0.17 (0.41)	0.22 (0.46)	0.22 (0.49)	0.28 (0.50)	0.38 (0.63)	0.32 (0.56)	0.37 (0.61)	0.44 (0.66)	0.27 (0.52)
3. Felt tired	0.47 (0.59)	0.51 (0.60)	0.59 (0.62)	0.67 (0.67)	0.85 (0.66)	0.92 (0.76)	0.74 (0.70)	0.88 (0.68)	0.96 (0.76)	0.85 (0.73)
4. Restless	0.59 (0.65)	0.53 (0.61)	0.63 (0.65)	0.59 (0.66)	0.75 (0.66)	0.76 (0.71)	0.60 (0.66)	0.64 (0.66)	0.78 (0.73)	0.55 (0.68)
5. I was no good anymore	0.19 (0.46)	0.17 (0.44)	0.26 (0.53)	0.33 (0.60)	0.35 (0.58)	0.43 (0.68)	0.38 (0.62)	0.40 (0.65)	0.44 (0.69)	0.50 (0.72)
6. Cried a lot	0.20 (0.46)	0.19 (0.46)	0.24 (0.51)	0.37 (0.64)	0.39 (0.63)	0.41 (0.66)	0.28 (0.55)	0.32 (0.58)	0.36 (0.64)	0.38 (0.64)
7. Hard to concentrate	0.62 (0.61)	0.55 (0.63)	0.68 (0.65)	0.68 (0.68)	0.90 (0.65)	0.74 (0.71)	0.60 (0.66)	0.69 (0.66)	0.82 (0.72)	0.70 (0.70)
8. I hated myself	0.16 (0.43)	0.16 (0.44)	0.23 (0.52)	0.31 (0.61)	0.30 (0.56)	0.28 (0.59)	0.26 (0.55)	0.28 (0.57)	0.33 (0.63)	0.41 (0.68)
9. I was a bad person	0.12 (0.34)	0.16 (0.41)	0.23 (0.48)	0.30 (0.58)	0.34 (0.56)	0.28 (0.56)	0.24 (0.51)	0.25 (0.54)	0.30 (0.60)	0.37 (0.64)
10. Felt lonely	0.36 (0.55)	0.28 (0.53)	0.38 (0.60)	0.55 (0.71)	0.59 (0.68)	0.65 (0.74)	0.59 (0.70)	0.60 (0.70)	0.64 (0.74)	0.65 (0.75)
11. Nobody really loved me	0.14 (0.41)	0.14 (0.42)	0.21 (0.50)	0.29 (0.59)	0.29 (0.55)	0.23 (0.54)	0.18 (0.48)	0.20 (0.50)	0.22 (0.52)	0.30 (0.61)
12. Not as good as other kids	0.30 (0.53)	0.25 (0.51)	0.33 (0.59)	0.36 (0.63)	0.37 (0.59)	0.42 (0.66)	0.40 (0.64)	0.47 (0.68)	0.53 (0.70)	0.56 (0.74)
13. Did everything wrong	0.16 (0.40)	0.17 (0.42)	0.22 (0.49)	0.30 (0.58)	0.33 (0.56)	0.35 (0.62)	0.31 (0.57)	0.28 (0.54)	0.30 (0.56)	0.46 (0.70)
Composite score	4.04 (3.51)	3.97 (3.86)	4.92 (4.49)	5.91 (5.64)	6.59 (5.25)	6.83 (5.93)	5.70 (5.58)	6.21 (5.50)	7.03 (6.06)	6.89 (6.41)
Cronbach's alpha (α)	.80	.84	.87	.91	.91	.91	.92	.91	.91	.92
Omega total (ω_t)	.83	.87	.89	.92	.92	.93	.93	.92	.93	.94

<i>Skewness/Kurtosis</i>	Wave 1 (<i>N</i> = 7,364)	Wave 2 (<i>N</i> = 6,716)	Wave 3 (<i>N</i> = 6,019)	Wave 4 (<i>N</i> = 4,997)	Wave 5 (<i>N</i> = 4,497)	Wave 6 (<i>N</i> = 3,335)	Wave 7 (<i>N</i> = 3,305)	Wave 8 (<i>N</i> = 3,856)	Wave 9 (<i>N</i> = 3,915)	Wave 10 (<i>N</i> = 4,016)
1. Miserable/unhappy	0.43/-0.68	0.33/-0.66	0.27/-0.64	0.10/-0.99	0.15/-0.62	0.04/-1.11	0.25/-0.75	0.21/-0.77	0.14/-0.92	0.24/-1.02
2. Did not enjoy anything	2.55/6.02	2.36/4.97	1.93/2.93	2.17/3.91	1.54/1.44	1.39/0.75	1.56/1.44	1.43/0.91	1.19/0.18	1.84/2.48
3. Felt tired	0.84/-0.29	0.71/-0.46	0.55/-0.62	0.50/-0.77	0.18/-0.76	0.14/-1.25	0.41/-0.91	0.16/-0.88	0.07/-1.25	0.23/-1.10
4. Restless	0.66/-0.59	0.71/-0.47	0.54/-0.68	0.68/-0.61	0.32/-0.76	0.39/-0.98	0.67/-0.62	0.55/-0.72	0.37/-1.06	0.84/-0.46
5. I was no good anymore	2.43/5.22	2.58/6.08	1.98/2.96	1.64/1.53	1.39/0.92	1.31/0.33	1.42/0.86	1.37/0.61	1.26/0.18	1.07/-0.27
6. Cried a lot	2.20/4.18	2.41/5.14	2.10/3.53	1.50/0.97	1.35/0.65	1.37/0.54	1.87/2.50	1.67/1.71	1.54/1.08	1.45/0.84
7. Hard to concentrate	0.43/-0.66	0.70/-0.50	0.45/-0.73	0.51/-0.80	0.10/-0.66	0.41/-0.96	0.65/-0.63	0.45/-0.76	0.28/-1.05	0.49/-0.89
8. I hated myself	2.77/7.22	2.83/7.42	2.25/4.10	1.80/1.96	1.76/2.05	1.97/2.64	2.01/3.02	1.94/2.62	1.71/1.59	1.40/0.54
9. I was a bad person	2.98/8.58	2.46/5.52	2.06/3.46	1.81/2.15	1.46/1.14	1.92/2.61	2.02/3.22	2.05/3.22	1.81/2.06	1.50/0.97
10. Felt lonely	1.21/0.49	1.70/2.00	1.35/0.74	0.90/-0.50	0.71/-0.63	0.66/-0.91	0.78/-0.64	0.73/-0.69	0.67/-0.88	0.69/-0.94
11. Nobody really loved me	2.93/8.26	3.00/8.58	2.34/4.62	1.93/2.49	1.74/2.03	2.28/4.09	2.68/6.30	2.49/5.33	2.37/4.59	1.90/2.27
12. Not as good as other kids	1.57/1.53	1.96/2.98	1.59/1.45	1.56/1.17	1.34/0.75	1.30/0.42	1.36/0.62	1.14/0.01	0.96/-0.39	0.89/-0.63
13. Did everything wrong	2.28/4.44	2.42/5.31	2.11/3.70	1.77/1.99	1.48/1.22	1.56/1.20	1.70/1.83	1.82/2.37	1.72/1.97	1.19/0.01

Note. SMFQ = Short Mood and Feelings Questionnaire.

Figure 1
Schematic Presentation of the Longitudinal Factor Model



Note. For visual presentation, we only depict age 11 (wave 1) and age 13 (wave 2).

Parameter Estimates

Table 2 depicts freely CFA estimated factor loadings and thresholds. Factor loadings were all within an acceptable range from 0.34 to 0.94. The maximum difference in item loadings across the scale within one wave was observed at Age 11, a difference in standardized loadings of 0.50. Across waves, the items *I was not good anymore* and *I hated myself* had the highest factor loadings and the items *felt tired* and *restless* the lowest factor loadings. By each item, deviations of factor loadings across waves ranged from 0.08 to 0.40. The items *did not enjoy anything*, *felt tired*, *restless*, and *I was a bad person* displayed the highest deviations over time (highest deviation between all timepoints ≥ 0.30 for all these items, see Table 2). For the majority of items, the lowest loading for the same item over time was found at Wave 1 (Age 11, for 11/13 items), and the next lowest loading was at Wave 2 or Wave 3 (Age 13 or 14, for 8/13 items). The items *I was no good anymore*, *I hated myself*, and *nobody really loved me* had the highest item thresholds over time. The symptoms with the lowest item thresholds were *miserable/unhappy*, *felt tired*, *restless*, and *hard to concentrate*.

Research Aim 1: Longitudinal MI

MI Testing

Ages 11–26 (all waves): Configural model fit over time for Waves 1–10 was good as indicated by CFI and RMSEA (Table 3). When factor loadings and item thresholds were constrained to be equal the overall scalar model fit was good, Δ RMSEA values supported invariance, but Δ CFI indicated that model fit deteriorated substantially. Despite this mixed support of scalar invariance, we explored strict MI: Overall model fit was good and Δ RMSEA and Δ CFI did

not deteriorate compared to the scalar MI model. Because we found a violation of longitudinal scalar MI, we aimed to establish partial MI across time by iteratively relaxing constraints on the symptoms with the highest deviation in loadings and/or thresholds across time. However, even if we freed the factor loadings and thresholds for the four symptoms *did not enjoy anything*, *felt tired*, *restless*, and *I was a bad person*, the CFI could only be improved to .967. Hence, Δ CFI still was just above the cut-off required for MI (.013). We set no more parameters free because we found no other obvious candidate items (next highest deviation was .05).

Adolescent models, ages 11–18 (Waves 1–5): Findings mirror those from models with all waves: Configural model fit over time was good according to CFI and RMSEA. While overall scalar model fit was good, Δ RMSEA values supported invariance, but Δ CFI indicated that model fit deteriorated substantially. The strict MI model did not deviate from the scalar MI model. Again, we relaxed constraints on the symptoms with the highest deviation in loadings and/or thresholds across time to establish partial MI. When we freed the factor loadings and thresholds for the four symptoms *did not enjoy anything*, *felt tired*, *restless*, and *I was a bad person*, the CFI improved to .969 and Δ CFI was thus still above the cut-off required for MI (.011). As above, no more parameters were set free.

Emerging adult models, ages 19–26 (Waves 6–10): We found good fit for all models across time (Table 3). Moreover, model fit did not substantially decrease when constraints were added, and we could confirm strict longitudinal factor MI.

Further exploratory tests of longitudinal MI: As noted above, lower loadings occurred in the first three waves, suggesting a lack of invariance over the full age range. We thus performed further MI testing across different waves to parameterize at which ages SMFQ is longitudinally invariant. As the first wave showed the lowest loadings

Table 2
Freely Estimated Standardized Factor Loadings and Item Thresholds From Ages 11 to 26 (Waves 1–10)

Freely estimated factor loadings (λ)	λ_{k1}	λ_{k2}	λ_{k3}	λ_{k4}	λ_{k5}	λ_{k6}	λ_{k7}	λ_{k8}	λ_{k9}	λ_{k10}	$D_{\max}$
1. Miserable/unhappy	.60	.66	.72	.77	.75	.76	.79	.81	.80	.75	.21
2. Did not enjoy anything	.44	.52	.57	.74	.73	.77	.82	.83	.84	.74	.40
3. Felt tired	.39	.49	.47	.56	.52	.55	.69	.56	.60	.63	.30
4. Restless	.34	.47	.44	.56	.51	.57	.63	.52	.61	.65	.31
5. I was no good anymore	.84	.86	.89	.90	.89	.91	.92	.91	.94	.93	.10
6. Cried a lot	.62	.72	.71	.68	.68	.68	.68	.64	.64	.66	.10
7. Hard to concentrate	.50	.63	.59	.60	.59	.66	.76	.67	.69	.75	.26
8. I hated myself	.82	.89	.89	.89	.87	.92	.94	.93	.94	.93	.12
9. I was a bad person	.52	.62	.67	.83	.80	.81	.85	.84	.87	.88	.36
10. Felt lonely	.76	.78	.81	.81	.80	.73	.76	.71	.73	.76	.10
11. Nobody really loved me	.84	.85	.85	.85	.85	.84	.82	.79	.82	.87	.08
12. Not as good as other kids	.73	.79	.80	.84	.84	.86	.88	.81	.83	.88	.15
13. Did everything wrong	.71	.77	.85	.80	.89	.90	.91	.83	.85	.93	.22

Freely estimated thresholds (τ)	$\tau_{k11/12}$	$\tau_{k21/12}$	$\tau_{k31/12}$	$\tau_{k41/12}$	$\tau_{k51/12}$	$\tau_{k61/12}$	$\tau_{k71/12}$	$\tau_{k81/12}$	$\tau_{k91/12}$	$\tau_{k101/12}$
1. Miserable/unhappy	−0.15/2.04	−0.42/1.80	−0.52/1.94	−0.92/1.30	−0.78/1.69	−0.94/1.04	−0.67/1.92	−0.81/1.83	−0.95/1.52	−0.63/1.34
2. Did not enjoy anything	1.32/2.70	1.23/2.76	1.24/2.77	1.39/2.75	1.03/3.04	0.86/2.25	1.03/2.81	1.06/2.80	0.69/2.54	1.11/2.78
3. Felt tired	0.26/1.85	0.16/1.91	−0.03/1.70	−0.18/1.54	−0.56/1.26	−0.50/0.88	−0.28/1.48	−0.56/1.20	−0.49/0.91	−0.46/1.11
4. Restless	0.03/1.55	0.10/1.72	−0.09/1.52	0.02/1.58	−0.33/1.41	−0.18/1.28	0.05/1.78	−0.01/1.63	−0.21/1.29	0.29/1.70
5. I was no good anymore	1.91/3.66	2.06/3.84	1.76/3.64	1.53/3.58	1.29/3.90	1.27/3.19	1.36/3.67	1.39/3.44	1.35/3.81	0.95/3.12
6. Cried a lot	1.18/2.55	1.37/2.75	1.18/2.43	0.89/2.01	0.69/2.01	0.66/1.68	1.03/2.26	0.90/2.05	0.74/1.80	0.76/1.77
7. Hard to concentrate	−0.09/1.90	0.07/1.98	−0.20/1.62	−0.20/1.50	−0.74/1.23	−0.27/1.32	−0.02/2.04	−0.25/1.71	−0.46/1.34	−0.26/1.84
8. I hated myself	2.05/3.69	2.48/4.33	2.08/3.77	1.75/3.38	1.62/3.52	2.21/3.97	2.42/4.64	2.22/4.38	2.21/4.17	1.65/3.33
9. I was a bad person	1.55/3.12	1.40/2.91	1.22/2.54	1.33/2.56	1.04/2.98	1.43/2.84	1.60/3.27	1.65/3.33	1.58/3.06	1.41/2.96
10. Felt lonely	0.67/2.72	1.02/2.86	0.76/2.54	0.33/2.04	0.11/2.10	0.09/1.51	0.21/1.89	0.11/1.67	0.07/1.51	0.09/1.53
11. Nobody really loved me	2.22/3.89	2.30/3.80	1.87/3.35	1.60/2.95	1.45/3.29	1.72/3.05	1.97/3.13	1.77/2.90	1.87/3.13	1.73/2.92
12. Not as good as other kids	1.05/2.84	1.35/2.96	1.09/2.67	1.19/2.72	0.99/2.97	1.00/2.62	1.07/2.98	0.57/2.16	0.50/2.15	0.51/2.28
13. Did everything wrong	1.64/3.35	1.61/3.34	1.75/3.68	1.80/3.77	1.41/3.84	1.56/3.50	1.75/4.03	1.37/3.22	1.35/3.38	1.14/3.28

Note. K refers to the number of the item. τ_1 and τ_2 constitute the first and second threshold, respectively. $D_{\max}$ = maximal deviation of factor loadings across waves.

Table 3
Model Fit for MI and Sum Score Models

Model	Waves 1–10					Adolescent model (Waves 1–5)					Emerging adult model (Waves 6–10)				
	$\chi^2(df)$	CFI	RMSEA	ΔCFI	$\Delta RMSEA$	$\chi^2(df)$	CFI	RMSEA	ΔCFI	$\Delta RMSEA$	$\chi^2(df)$	CFI	RMSEA	ΔCFI	$\Delta RMSEA$
Configural MI	9,345 (7,625)	.980	.014			4,358 (1,875)	.980	.021			3,941 (1,875)	.972	.029		
Scalar MI	11,589 (7,849)	.965	.020	.015	.006	6,968 (1,975)	.965	.026	.015	.005	4,961 (1,975)	.966	.031	.006	.002
Strict MI	11,947 (7,625)	.964	.020	.001 ^a	.000	8,503 (2,027)	.962	.027	.003 ^a	.001	5,633 (2,027)	.968	.030	.002	.001
Sum Score	31,211 (7,745)	.956	.057	.024^b	.043^b	25,197 (1,935)	.935	.068	.045^b	.047^b	18,592 (1,935)	.962	.083	.010^b	.054^b
Waves 1–4															
Configural MI	3,383 (1,190)	.980	.022			7,943 (6,165)	.979	.013			6,840 (4,860)	.975	.019		
Scalar MI	4,781 (1,265)	.971	.026	.009	.004	9,675 (6,364)	.965	.016	.014	.003	8,146 (5,035)	.969	.021	.006	.002
Strict MI	5,976 (1,304)	.968	.027	.003	.001	11,939 (6,468)	.960	.018	.005 ^a	.002	9,799 (5,125)	.969	.021	.000	.000
Sum Score	22,919 (1,238)	.930	.071	.050^b	.046^b	28,701 (6,273)	.953	.049	.026^b	.036^b	26,571 (4,956)	.956	.068	.019^b	.049^b

Note. MI = measurement invariance; CFI = comparative fit index; RMSEA = root-mean-square error of approximation; df = degrees of freedom. Available data for models: adolescent data for models; adolescent model (61%), Emerging adult model (81%), Waves 1–3 (79%), Waves 1–4 (65%), Waves 3–10 (55%). $\Delta CFI \geq .010$ and $\Delta RMSEA \geq .007$ indicate substantial deterioration in model fit (Neufeld et al., 2022). Models are compared with the prior model consisting of one less level of constraints. Bolded numbers indicate noninvariance.

^a These comparisons are not meaningful according to our criteria to establish MI. This is because MI was not established in the step before so that we technically cannot compare this model with the next model. However, we report these values in case researchers apply less strict criteria for their model comparisons. ^b Except for the sum score models, which are compared to the configural model.

compared to all other waves, we first tested MI for Waves 2–10 (13–26 years). Results mirrored those of Waves 1–10: Configural MI was good, $\Delta RMSEA$ supported scalar invariance, but ΔCFI did not (Table 3). Model fit for strict MI was still good. Freeing constraints on the symptoms *did not enjoy anything*, *felt tired*, *restless*, and *I was a bad person*, did not lead to partial MI (ΔCFI was still at least .012). We then tested Waves 3–10 (ages 14–26 years): Model fit was good for all models and strict factor MI across time could be confirmed (Table 3). As an additional time-interval, we chose Waves 1–4 (ages 11–17). For this model, we observed the same pattern: model fit was good and full strict MI could be confirmed across time.

Research Aim 2: Sex MI

MI Across Sex

For sex, model fit for all models was good according to CFI and good or acceptable according to RMSEA (Supplemental Table S4). Strict MI across sex could be confirmed for ages 13 (Wave 2), 16 (Wave 4), 18 (Wave 5), 19 (Wave 6), 23 (Wave 8), 24 (Wave 9), and 26 (Wave 10). At Age 11 (Wave 1), scalar invariance was not indicated according to ΔCFI . Freeing factor loadings of the item *I cried a lot* ($\lambda_{\text{male}}: .59$, $\lambda_{\text{female}}: .67$) and thresholds of *feeling restless* ($\tau_{1\text{male}}: -0.08$, $\tau_{2\text{male}}: 1.22$; $\tau_{1\text{female}}: 0.10$, $\tau_{2\text{female}}: 1.47$) led to partial scalar MI and partial strict MI. At Age 14 (Wave 3) and Age 22 (Wave 7), scalar MI was not supported by $\Delta RMSEA$. In both cases, freeing the factor loadings of the item *I cried a lot* ($\lambda_{\text{male}}: .68$, $\lambda_{\text{female}}: .75$, 14 years; $\lambda_{\text{male}}: .61$, $\lambda_{\text{female}}: .73$, 22 years) led to partial scalar MI and also partial strict MI.

Research Aim 3: Appropriateness of Sum Scores and Factor Scores

Model Fit

The equal loading sum score models had acceptable or good fit, yet model fit deteriorated when compared to the unconstrained models (Table 3).

Reliability as measured by Cronbach's α and ω total was always higher when unequal loadings were considered (ω_{total}) compared with when equal loadings were assumed (α). However, this difference was only minimal (.01–.03), and all values for both coefficients were at least good for all waves (Waves 1–3 good: [$> .80$ but $< .90$], Waves 4–10 excellent [$> .90$]; McNeish, 2018, see Table 1).

External Validation

Table 4 shows associations with external validators for the sum scores as well as extracted factor scores from models specified with minimal (configural) versus maximal (strict) longitudinal MI. Despite not being invariant by all indicators at the scalar level, we used the Waves 1–10 model with the highest level of restrictions (strict MI) because the fit of this model was still good, and only the ΔCFI indicated scalar noninvariance. Extracted factor scores (based on configural vs. strict MI) were almost identical in their associations with external validators (the highest deviation was .01); these associations were also very similar for sum scores. Differences in association with external constructs between sum scores and factor scores did not reach a meaningful effect size (ranging from .00 to .08 [configural/strict MI], see Table 4), and the respective confidence intervals

Table 4
External Variable Correlations With Sum Scores/Factor Scores Derived From Configural MI Model/Factor Scores Derived From Strict MI Model

Wave (age)	Sex ^c (male = 1)	Psychotic 17 years	Anxiety 21 years	Health 19 years	Health 21 years	Health 22 years	SDQ ^a 12 years	SDQE ^a 12 years	SDQ ^a 13 years	SDQE ^a 13 years	LE 12 years
Wave 1 (11 years)	.04 ^b /.00 ^b /.00 ^b	.16/.17/.17	.16/.22/.22	.13/.17/.17	.13/.16/.16	.16/.18/.18	.29/.29/.29	.20/.22/.22	.28/.31/.31	.20/.23/.23	.12/.14/.14
Wave 2 (13 years)	-.10 ^b /.14 ^b /.14	.22/.24/.24	.24/.28/.28	.19/.22/.22	.17/.21/.21	.15/.18/.18	.17/.21/.22	.16/.18/.18	.23/.28/.28	.21/.24/.24	.18/.19/.19
Wave 3 (14 years)	-.18 ^b /.20 ^b /.20	.27/.31/.31	.27/.31/.31	.23/.28/.28	.19/.24/.23	.17/.21/.21	.17/.21/.21	.15/.18/.18	.21/.25/.25	.21/.24/.24	.15/.20/.20
Wave 4 (17 years)	-.24 ^b /.25 ^b /.25	.34/.33/.33	.38/.38/.37	.28/.32/.31	.25/.26/.26	.22/.25/.25	.16/.19/.19	.18/.18/.18	.18/.22/.21	.21/.22/.22	.22/.23/.23
Wave 5 (18 years)	-.15 ^b /.17 ^b /.17	.26/.26/.26	.37/.39/.39	.29/.31/.31	.24/.27/.27	.20/.24/.24	.18/.19/.19	.17/.17/.17	.20/.21/.21	.19/.21/.21	.13/.16/.17
Wave 6 (19 years)	-.18 ^b /.15 ^b /.15	.24/.21/.21	.36/.44/.44	.34/.38/.38	.24/.29/.29	.21/.23/.22	.18/.20/.20	.20/.22/.22	.21/.21/.21	.23/.24/.24	.14/.14/.14
Wave 7 (22 years)	-.11 ^b /.13 ^b /.13	.24/.23/.23	.72/.70/.70	.28/.32/.32	.37/.40/.39	.33/.33/.33	.18/.19/.19	.19/.22/.22	.20/.22/.22	.20/.24/.24	.13/.19/.20
Wave 8 (23 years)	-.11 ^b /.13 ^b /.13	.20/.25/.25	.52/.57/.57	.25/.31/.31	.31/.35/.35	.34/.34/.34	.16/.20/.20	.16/.21/.21	.20/.23/.23	.19/.22/.22	.10/.16/.16
Wave 9 (24 years)	-.10 ^b /.12 ^b /.12	.19/.21/.21	.50/.55/.55	.27/.30/.30	.30/.33/.33	.30/.33/.32	.17/.20/.20	.18/.20/.20	.20/.22/.22	.19/.21/.21	.11/.15/.15
Wave10 (26 years)	-.16 ^b /.13 ^b /.12	.22/.23/.23	.46/.51/.51	.27/.29/.29	.29/.33/.33	.29/.32/.31	.18/.22/.22	.18/.21/.21	.21/.21/.21	.18/.21/.21	.14/.18/.18

Note. SDQ = Strengths and Difficulties Questionnaire; LE = Life events; MI = measurement invariance; SDQE = Strengths and Difficulties Questionnaire Emotional subscale; SMFQ = Short Mood and Feelings Questionnaire. Bolded indicates concurrent associations or the closest to the concurrent associations (some external variables were obtained in between waves where SMFQ was obtained).
^a SDQ rated by the mother. SDQE refers to the emotional subscale of the SDQ. ^b Nonsignificant; all other p values < .001. ^c Based on point biserial correlation, whereas the other correlations represent Pearson correlations.

overlapped (Supplemental Table S5). Thus, associations with external validators are outlined below for factor and sum scores at once.

At Wave 1, there were no significant sex differences, whereas from Waves 2 to 10 males showed lower depressive scores, with point biserial correlations ranging from $-.10$ to $-.25$ ($p < .001$). Across waves, SMFQ scores were significantly associated with psychotic symptoms at Age 17 (Pearson's $r = .16-.34$), anxiety symptoms at Age 22 ($r = .16-.72$), life events at Age 17 ($r = .10-.23$), self-rated health at ages 19, 21, and 22 ($r = .13-.40$) and the SDQ mother report ages 12 and 13 ($r = .16-.31$).

Contemporaneous correlations between factor scores and sum scores ranged from .92 (Wave 2 correlation between sum scores and configural factor scores) to .95 (Wave 10 correlation between sum scores and both types of factor scores; Supplemental Table S6). This shows that the sum score model explains between 85% and 90% of the variance in the more precise factor model. Contemporaneous correlations for factor scores of the configural versus strict MI models ranged from .94 (Waves 2–4) to .99 (Waves 9 and 10), explaining between 88% and 98% of the variance.

Discussion

We examined MI of the SMFQ across time and sex from childhood to emerging adulthood (ages 11–26 years) in a population-based study in the South-West of England. The one-factor structure of the SMFQ displayed good model fit across all waves, in line with previous psychometric studies on the SMFQ (Thabrew et al., 2018; Turner et al., 2014). However, strict factor MI was only confirmed from ages 14 to 26 (Waves 3–10) and ages 11 to 17 (Waves 1–4) but not fully from ages 11 to 26 (Wave 1–10). These findings align with studies that report a lower discrimination of the SMFQ against clinical interview in younger people aged 11–13 (Rhew et al., 2010) compared to ages above 13 years (Thabrew et al., 2018), and lower factor loadings in younger boys compared to older boys (Messer et al., 1995).

Research Aim 1: Longitudinal MI

The present study elucidated different functioning of the SMFQ at ages 11 and 13 (Waves 1 and 2); longitudinal scalar invariance was not met when these ages were included with the rest of the waves. However, when not including these ages, we established the highest level of MI in ages 14–26 years (Waves 3–10). Differences in loadings and thresholds of MFQ items at Age 11–13 compared with the other ages may be attributable to a different presentation and expression of depressive symptoms in these age groups. As mood disorder symptoms increase from Age 10 onwards (Solmi et al., 2021), specific symptoms may fluctuate more during these ages as reflected in lower depression rank-order stability over time (Cole & Martin, 2005). Our findings indicate that some caution is required when comparing SMFQ scores across ages 11–13 with scores at later ages, as some degree of bias may be introduced. We therefore recommend independent testing of MI when comparing SMFQ scores from ages 11 to 13 to scores at later ages to ensure accurate scoring for examining developmental changes. One could, however, argue that the violations of MI from ages 11 to 18 may, however, not be of large practical significance. Item factor loadings were acceptable for ages 11 and 13, and model fit was still good for all models. We tested different a priori as well as exploratory models to pinpoint the exact age range across

which MI holds. The SMFQ demonstrated strict MI from ages 14 to 26 (Waves 3–10) and ages 11 to 17 (Waves 1–4) and may thus be used to investigate depressive symptom development across these age ranges.

In the broader literature, the Social Behavior Questionnaire has shown a similar pattern and functioned psychometrically well across ages 11–17 with minor violations of MI in younger ages (mainly ages 11 and 13; Murray et al., 2019). Specifically, the residual variances of symptoms were found to be higher in younger ages. The authors suggested that with increasing age, participants may have an enhanced ability to report symptoms or became more attentive toward symptoms, which we concur may also be relevant for the SMFQ.

Noninvariant Items

The symptoms *felt tired*, *restless*, and *hard to concentrate* had the lowest factor loadings, especially at ages 11 and 13. These somatic and cognitive symptoms may be more normative during these ages, and could reflect the difficulty some children have adjusting to school demands. As the ability to identify and differentiate emotions increases from childhood to young adulthood (Gur et al., 2012), these symptoms may be less reflective of the depression construct in younger compared to older ages. Accordingly, these items may not be ideally suited to reveal differences between individuals on the depression construct in early adolescence. Internal feelings of restlessness manifest during adolescence and are more apparent in overt forms of behavior in childhood (Weyandt et al., 2003). Therefore, the internal component of feeling restless may be more strongly related to the underlying depression construct at later developmental stages. Indeed, items assessing restlessness and concentration problems served as excellent indicators for a latent attention deficit hyperactivity disorder (ADHD) construct in ages 13–17 (Murray et al., 2019). It could thus be that such items may not ideally capture the distinct aspects of the inner states of depression and instead reflect more overt behavior problems related to attention/hyperactivity at younger ages. As ADHD and depression symptoms are often related (for a meta-analysis see Meinzer et al., 2014), these items still contributed meaningfully to the depression construct in the present study. However, given that these items had the lowest thresholds it may be difficult to discriminate severely depressed individuals in clinical studies when these items are included. Lower factor loadings of *I did not enjoy anything* in the first two waves could be attributable to anhedonia's varying presentation as a function of age. In previous work, anhedonia was more frequently reported during adulthood than adolescence (Rice et al., 2019). In our study, the first threshold for this item was higher in earlier waves. This indicates that in younger adolescents, anhedonia reflects a higher level of severity on the depression construct, making it less likely that younger adolescents will endorse this item.

Research Aim 2: MI Across Sex

We could not identify systematic sex bias in the SMFQ across waves, based on the binary parent reports of sex (male/female). While there are limitations to this measure of sex which does not encompass gender more broadly, only a small percentage of young people identify as nonbinary and transgender (population estimates

range from 2% to 5%, Richards et al., 2016), so this is unlikely to have altered findings. While future research should be more inclusive, the binary reporting of sex at baseline is typical in many longitudinal studies, especially for those which started many years ago (ALSPAC baseline was from 1991 to 1992, Boyd et al., 2013).

Some small violations of MI across sex were only apparent for the scalar MI models at ages 11 (Wave 1), 14 (Wave 3), and 22 (Wave 7), and noninvariance was only indicated by failing to meet one of two cut-off criteria, not both (Neufeld et al., 2022). In these waves, the thresholds of the symptom *feeling restless* were lower in males than in females, which could align with our discussion of ADHD symptoms above, which are more prevalent among boys than girls (Meinzer et al., 2014). Also in the three mentioned waves, females showed higher factor loadings for the symptom *I cried a lot*. In a prior SMFQ study in 17-year-old Norwegians, females had lower thresholds of reporting this item than males (Lundervold et al., 2013). However, the differences in the present study were not large and factor loadings for this item were also good in males. Furthermore, setting this loading to be equal across sex in the other waves did not lead to a decrement in model fit. The highest levels of MI could be established after relaxing constraints on these parameters at these three waves, adding evidence of low levels of sex bias in the SMFQ. As partial MI results are often sample-dependent, we would not recommend applying them in different samples given that they were inconsistent across time and negligible in their magnitude. We therefore do not expect any systematic bias in the scores between females and males indicating that SMFQ scores can be used to approximate differences between sexes. Based on the association of sex with SMFQ scores, we found the expected sex differences from ages 13 to 26 (Salk et al., 2017; i.e., females displayed higher depression scores than males) but not at Age 11. This may be because depression onset is typically between ages 12 and 15 when puberty begins, during which females show a steeper increase of symptoms than males (Patalay & Fitzsimons, 2018; Shore et al., 2018). In interpreting these results, it is important to consider that male participants were more likely to drop out of the study, in line with other panel studies (Satherley et al., 2015). The reasons for this pattern and potential consequences for MI across sex needs to be investigated in subsequent work.

Research Aim 3: Appropriateness of Sum Scores and Factor Scores

Despite substantial variability of indicator loadings, models with equality constraints on factor loadings showed acceptable model fit. Likewise, the reliability estimates of both sum scores and differential weights were good-to-excellent across all waves, and virtually indistinguishable from each other (.03 or less). The consistency in reliability estimates is particularly notable, given that within one wave, standardized item loadings differed by up to 0.50, equivalent to a medium effect size. All scores showed the expected (and virtually indistinguishable) associations with external constructs. This indicates that although sum score models fit the data worse than the factor score models because they do not capture the differential impact of symptoms, this bias may not be of large practical significance, in line with recent considerations (Widaman & Revelle, 2022). Furthermore, sum score and factor score estimates were extremely similar (contemporaneous correlations ranged from 0.92 to 0.95). Given the cross-study consistency in the SMFQ single-factor structure

(e.g., present study, Thabrew et al., 2018; Turner et al., 2014) and our finding that moderate loading differences across items did not appreciably influence reliability and validity of the sum scores, it is likely that use of the SMFQ sum scores will be appropriate in most samples. However, researchers should consider testing the appropriateness of the SMFQ sum scores in their own data (McNeish & Wolf, 2020; Widaman & Revelle, 2022), particularly if their sample is considerably different than those cited above. Regardless, it is important to note that use of sum scores leads to a consistent use of the scale across studies, enabling more comparable results across studies (McNeish & Wolf, 2020; Widaman & Revelle, 2022).

It is important to mention that factor scores derived using the empirical Bayes method (Muthén, 1998–2004) based on the MI models with the lowest restriction and the highest restriction led to nearly identical results when correlated with external variables or sum scores. This was the case despite deriving factor scores with strict MI assumptions even though scalar MI was not established according to ΔCFI in the Wave 1–10 model. Although we did not test external associations using the full factor model, others have found such a model to yield comparable external associations to the sum score model (Widaman & Revelle, 2022). Overall, the SMFQ seems to accurately capture the underlying depression construct and exhibit consistent external associations regardless of the scoring methods used.

Limitations

First, for categorical data, there is no clear consensus on how to evaluate violations of MI and how consequential the violations are (Liu et al., 2017). The performance of fit indices using categorical indicators has not been thoroughly examined (Neufeld et al., 2022). However, external associations do not seem to be affected by a small degree of noninvariance, as we observed nearly identical external associations with factor scores from the models employing the least versus highest invariance constraints. Still, simulation studies are needed to further elucidate the consequences of such small violations of MI. Second, external measures of mental health were limited by the study design because the external measures were assessed at different timepoints. However, this analysis was primarily intended to assess the relative external validity of sum scores versus various factor scores as opposed to comprehensively assessing associations over time with the SMFQ. Third, despite having a reasonably large sample, we were unable to investigate MI across different ethnicities due to a lack of ethnic diversity in the sample. In a similar vein, the current findings only apply to the English version of the SMFQ. The SMFQ is available in multiple languages and has, for examples, been validated in Norwegian (Lundervold et al., 2013) and Spanish (Espada et al., 2022). However, language MI of SMFQ has yet to be tested; linguistic subtleties may change the meaning of some items. Fourth, we tested MI in an epidemiological study, which has implications for the interpretation of mean-level differences across time or sex on a population level. However, MI testing across clinical and nonclinical populations is still necessary. For example, SMFQ had lower discrimination in a depressed versus population sample when tested against clinical interviews in adolescents ages 11–13 (Rhew et al., 2010). Non-invariance of certain items may have contributed to this effect, which could affect comparison of the SMFQ in clinical versus

nonclinical populations. Fifth, some items that bear diagnostic relevance such as irritability are not captured by the SMFQ (Stringaris, 2011). It could be a strength of the SMFQ in terms of longitudinal MI that this item is not captured as it is prone to change developmentally (i.e., irritability may be more common in younger children). This way, researchers do not have to deal with items that change as a function of time. However, this could also be a drawback if this item captures core experiences of depression symptomatology in children and adolescents. Sixth, attrition was large and of concern. While we identified variables that were associated with missingness, unmeasured variables may have influenced attrition. Parameter estimates are therefore only unbiased under the assumption of MAR (White et al., 2016). Finally, at Waves 1, 2, 3, and 5 the SMFQ was administered as an interview in clinic, whereas at the other waves it was completed by post. While this could be a confounder, it was implicitly tested in the MI model (Waves 3–10 model was fully invariant and included both forms of administration).

Conclusion

The SMQF appears to measure the same underlying depression construct across different developmental stages from ages 14 to 26. Some caution is necessary when comparing the construct at ages 11–13 with ages greater than 17. SMFQ also appears strictly invariant across self-reported sex at most ages studied, with partial sex invariance found at only three waves. Further, the measure appears to be well-justified for use as a sum score, given its single-factor structure, good reliability when equal loadings were assumed, and comparable associations with external validators whether sum scores or factor scores were utilized. Both these findings support the use of sum scores as a practical choice in many settings including in clinical contexts. Overall, the SMFQ can be recommended for researchers interested in assessing change in depressive symptoms from early adolescence into emerging adulthood.

References

- Angold, A., Costello, E. J., Messer, S. C., & Pickles, A. (1995). Development of a short questionnaire for use in epidemiological studies of depression in children and adolescents. *International Journal of Methods in Psychiatric Research*, 5(4), 237–249.
- Anyan, F., & Hjemdal, O. (2018). Stress of home life and gender role socializations, family cohesion, and symptoms of anxiety and depression. *Women & Health*, 58(5), 548–564. <https://doi.org/10.1080/03630242.2017.1316343>
- Asparouhov, T., & Muthén, B. O. (2010). *Weighted least squares estimation with missing data. Technical appendix*. Muthén & Muthén.
- Boyd, A., Golding, J., Macleod, J., Lawlor, D. A., Fraser, A., Henderson, J., & Davey Smith, G. (2013). Cohort profile: The ‘children of the 90s’—The index offspring of the avon longitudinal study of parents and children. *International Journal of Epidemiology*, 42(1), 111–127. <https://doi.org/10.1093/ije/dys064>
- Chen, P. Y., Wu, W., Garnier-Villareal, M., Kite, B. A., & Jia, F. (2020). Testing measurement invariance with ordinal missing data: A comparison of estimators and missing data techniques. *Multivariate Behavioral Research*, 55(1), 87–101. <https://doi.org/10.1080/00273171.2019.1608799>
- Cole, D. A., & Martin, N. C. (2005). The longitudinal structure of the Children’s Depression Inventory: Testing a latent trait-state model. *Psychological Assessment*, 17(2), 144–155. <https://doi.org/10.1037/1040-3590.17.2.144>

- Coughlin, C., & Robins, R. W. (2017). Self-concept development. In A. Wenzel (Ed.), *SAGE encyclopedia of abnormal and clinical psychology* (pp. 1–6). Sage Publications.
- Cummings, C. M., Caporino, N. E., & Kendall, P. C. (2014). Comorbidity of anxiety and depression in children and adolescents: 20 years after. *Psychological Bulletin*, 140(3), 816–845. <https://doi.org/10.1037/a0034733>
- Dahl, R. E., Allen, N. B., Wilbrecht, L., & Suleiman, A. B. (2018). Importance of investing in adolescence from a developmental science perspective. *Nature*, 554(7693), 441–450. <https://doi.org/10.1038/nature25770>
- Edossa, A. K., Schroeders, U., Weinert, S., & Artelt, C. (2018). The development of emotional and behavioral self-regulation and their effects on academic achievement in childhood. *International Journal of Behavioral Development*, 42(2), 192–202. <https://doi.org/10.1177/0165025416687412>
- Espada, J. P., González, M. T., Fernández-Martínez, I., Orgilés, M., & Morales, A. (2022). Spanish validation of the short mood and feelings questionnaire (SMFQ) in children aged 8–12. *Psicothema*, 34(4), 610–620. <https://doi.org/10.7334/psicothema2022.54>
- Eyre, O., Jones, R. B., Agha, S. S., Wootton, R. E., Thapar, A. K., Stergiakouli, E., & Riegl, L. (2021). Validation of the short Mood and Feelings Questionnaire in young adulthood. *Journal of Affective Disorders*, 294, 883–888. <https://doi.org/10.1016/j.jad.2021.07.090>
- Finsas, M. C., Bufferd, S. J., Dougherty, L. R., Carlson, G. A., & Klein, D. N. (2018). Preschool psychiatric disorders: Homotypic and heterotypic continuity through middle childhood and early adolescence. *Psychological Medicine*, 48(13), 2159–2168. <https://doi.org/10.1017/S0033291717003646>
- Fraser, A., Macdonald-Wallis, C., Tilling, K., Boyd, A., Golding, J., Davey Smith, G., & Lawlor, D. A. (2013). Cohort profile: The avon longitudinal study of parents and children: ALSPAC mothers cohort. *International Journal of Epidemiology*, 42(1), 97–110. <https://doi.org/10.1093/ije/dys066>
- García-Cebrian, A., Gandhi, P., Demyttenaere, K., & Peveler, R. (2006). The association of depression and painful physical symptoms—a review of the European literature. *European Psychiatry*, 21(6), 379–388. <https://doi.org/10.1016/j.eurpsy.2005.12.003>
- Goodman, R. (2001). Psychometric properties of the strengths and difficulties questionnaire. *Journal of the American Academy of Child & Adolescent Psychiatry*, 40(11), 1337–1345. <https://doi.org/10.1097/00004583-200111000-00015>
- Gur, R. C., Richard, J., Calkins, M. E., Chiavacci, R., Hansen, J. A., Bilker, W. B., & Gur, R. E. (2012). Age group and sex differences in performance on a computerized neurocognitive battery in children age 8–21. *Neuropsychology*, 26(2), Article 251. <https://doi.org/10.1037/a0026712>
- Harris, P. A., Taylor, R., Thielke, R., Payne, J., Gonzalez, N., & Conde, J. G. (2009). Research electronic data capture (REDCap)—A metadata-driven methodology and workflow process for providing translational research informatics support. *Journal of Biomedical Informatics*, 42(2), 377–381. <https://doi.org/10.1016/j.jbi.2008.08.010>
- Hopkins, K., Crosland, P., Elliott, N., & Bewley, S. (2015). Diagnosis and management of depression in children and young people: Summary of updated NICE guidance. *The BMJ*, 350, Article h824. <https://doi.org/10.1136/bmj.h824>
- Horn, J. L. (1965). A rationale and test for the number of factors in factor analysis. *Psychometrika*, 32, 179–185. <https://doi.org/10.1007/BF02289447>
- Horwood, J., Salvi, G., Thomas, K., Duffy, L., Gunnell, D., Hollis, C., & Harrison, G. (2008). IQ and non-clinical psychotic symptoms in 12-year-olds: Results from the ALSPAC birth cohort. *The British Journal of Psychiatry*, 193(3), 185–191. <https://doi.org/10.1192/bjp.bp.108.051904>
- Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, 6(1), 1–55. <https://doi.org/10.1080/10705519909540118>
- Kingsbury, M., Weeks, M., MacKinnon, N., Evans, J., Mahedy, L., Dykxhoorn, J., & Colman, I. (2016). Stressful life events during pregnancy and offspring depression: Evidence from a prospective cohort study. *Journal of the American Academy of Child & Adolescent Psychiatry*, 55(8), 709–716. <https://doi.org/10.1016/j.jaac.2016.05.014>
- Korten, N. C., Comijs, H. C., Lamers, F., & Penninx, B. W. (2012). Early and late onset depression in young and middle aged adults: Differential symptomatology, characteristics and risk factors? *Journal of Affective Disorders*, 138(3), 259–267. <https://doi.org/10.1016/j.jad.2012.01.042>
- Kwong, A. S. (2019). Examining the longitudinal nature of depressive symptoms in the avon longitudinal study of parents and children (ALSPAC). *Wellcome Open Research*, 4, Article 126. <https://doi.org/10.12688/wellcomeopenres.15395.2>
- Lewis, G., Ioannidis, K., van Harmelen, A. L., Neufeld, S., Stochl, J., Lewis, G., & Goodyer, I. (2018). The association between pubertal status and depressive symptoms and diagnoses in adolescent females: A population-based cohort study. *PLOS ONE*, 13(6), Article e0198804. <https://doi.org/10.1371/journal.pone.0198804>
- Liu, Y., Millsap, R. E., West, S. G., Tein, J.-Y., Tanaka, R., & Grimm, K. J. (2017). Testing measurement invariance in longitudinal data with ordered-categorical measures. *Psychological Methods*, 22(3), 486–506. <https://doi.org/10.1037/met0000075>
- Löwe, B., Decker, O., Müller, S., Brähler, E., Schellberg, D., Herzog, W., & Herzberg, P. Y. (2008). Validation and standardization of the Generalized Anxiety Disorder Screener (GAD-7) in the general population. *Medical Care*, 46(3), 266–274. <https://doi.org/10.1097/MLR.0b013e318160d093>
- Lundervold, A. J., Breivik, K., Posserud, M. B., Stormark, K. M., & Hysing, M. (2013). Symptoms of depression as reported by Norwegian adolescents on the Short Mood and Feelings Questionnaire. *Frontiers in Psychology*, 4, Article 613. <https://doi.org/10.3389/fpsyg.2013.00613>
- McKenzie, D. P., Toumbourou, J. W., Forbes, A. B., Mackinnon, A. J., McMorris, B. J., Catalano, R. F., & Patton, G. C. (2011). Predicting future depression in adolescents using the Short Mood and Feelings Questionnaire: A two-nation study. *Journal of Affective Disorders*, 134(1–3), 151–159. <https://doi.org/10.1016/j.jad.2011.05.022>
- McNeish, D. (2018). Thanks coefficient alpha, we'll take it from here. *Psychological Methods*, 23(3), 412–433. <https://doi.org/10.1037/met0000144>
- McNeish, D., & Wolf, M. G. (2020). Thinking twice about sum scores. *Behavior Research Methods*, 52, 2287–2305. <https://doi.org/10.1037/met0000144>
- Meinzer, M. C., Pettit, J. W., & Viswesvaran, C. (2014). The co-occurrence of attention deficit/hyperactivity disorder and unipolar depression in children and adolescents: A meta-analytic review. *Clinical Psychology Review*, 34(8), 595–607. <https://doi.org/10.1016/j.cpr.2014.10.002>
- Messer, S. C., Angold, A., Costello, E. J., Loeber, R., Van Kammen, W., & Stouthamer-Loeber, M. (1995). Development of a short questionnaire for use in epidemiological studies of depression in children and adolescents: Factor composition and structure across development. *International Journal of Methods in Psychiatric Research*, 5, 251–262.
- Middleton, H., Shaw, I., Hull, S., & Feder, G. (2005). NICE guidelines for the management of depression. *The BMJ*, 330(7486), 267–268. <https://doi.org/10.1136/bmj.330.7486.267>
- Murray, A. L., Obsuth, I., Eisner, M., & Ribeaud, D. (2019). Evaluating longitudinal invariance in dimensions of mental health across adolescence: An analysis of the Social Behavior Questionnaire. *Assessment*, 26(7), 1234–1245. <https://doi.org/10.1177/1073191117721741>
- Muthén, B. O. (1998–2004). *Mplus technical appendices*. Muthén & Muthén.
- Muthén, L. K., & Muthén, B. O. (2017). *Mplus user's guide* (8th ed.).
- Neufeld, S. A. S., St Clair, M., Brodbeck, J., Wilkinson, P., Goodyer, I., & Jones, P. B. (2022, March 24). *Measurement invariance in longitudinal bifactor models: Review and application based on the p factor*. <https://osf.io/fbd3h>
- Neumann, A., van Lier, P. A., Gratz, K. L., & Koot, H. M. (2010). Multidimensional assessment of emotion regulation difficulties in adolescents using the difficulties in emotion regulation scale. *Assessment*, 17(1), 138–149. <https://doi.org/10.1177/1073191109349579>

- Nolen-Hoeksema, S., & Aldao, A. (2011). Gender and age differences in emotion regulation strategies and their relationship to depressive symptoms. *Personality and Individual Differences*, 51(6), 704–708. <https://doi.org/10.1016/j.paid.2011.06.012>
- Northstone, K., Lewcock, M., Groom, A., Boyd, A., Macleod, J., Timpson, N., & Wells, N. (2019). The avon longitudinal study of parents and children (ALSPAC): An update on the enrolled sample of index children in 2019. *Wellcome Open Research* 4, Article 51 <https://doi.org/10.12688/wellcomeopenres.15132.1>
- Patalay, P., & Fitzsimons, E. (2018). Development and predictors of mental ill-health and wellbeing from childhood to adolescence. *Social Psychiatry and Psychiatric Epidemiology*, 53(12), 1311–1323. <https://doi.org/10.1007/s00127-018-1604-0>
- R Core Team. (2021). *R: A language and environment for statistical computing* (Version. 4.0. 1). R Foundation for Statistical Computing.
- Rhew, I., Simpson, K., Tracy, M., Lymp, J., McCauley, E., Tsuang, D., & Vander Stoep, A. (2010). Criterion validity of the Short Mood and Feelings Questionnaire and one- and two-item depression screens in young adolescents. *Child and Adolescent Psychiatry and Mental Health*, 4, Article 8. <https://doi.org/10.1186/1753-2000-4-8>
- Rice, F., Riglin, L., Lomax, T., Souter, E., Potter, R., Smith, D. J., & Thapar, A. (2019). Adolescent and adult differences in major depression symptom profiles. *Journal of Affective disorders*, 243, 175–181. <https://doi.org/10.1016/j.jad.2018.09.015>
- Richards, C., Bouman, W. P., Seal, L., Barker, M. J., Nieder, T. O., & T'Sjoen, G. (2016). Non-binary or genderqueer genders. *International Review of Psychiatry*, 28(1), 95–102. <https://doi.org/10.3109/09540261.2015.1106446>
- Riglin, L., Leppert, B., Dardani, C., Thapar, A. K., Rice, F., O'Donovan, M. C., & Thapar, A. (2021). ADHD and depression: Investigating a causal explanation. *Psychological Medicine*, 51(11), 1890–1897. <https://doi.org/10.1017/S0033291720000665>
- Rossee, Y. (2012). Lavaan: An R package for structural equation modeling and more (Version 0.5–12 [BETA]). *Journal of Statistical Software*, 48(2), 1–36. <https://doi.org/10.18637/jss.v048.i02>
- Salk, R. H., Hyde, J. S., & Abramson, L. Y. (2017). Gender differences in depression in representative national samples: Meta-analyses of diagnoses and symptoms. *Psychological Bulletin*, 143(8), 783–822. <https://doi.org/10.1037/bul0000102>
- Santor, D. A., Ramsay, J. O., & Zuroff, D. C. (1994). Nonparametric item analyses of the Beck Depression Inventory: Evaluating gender item bias and response option weights. *Psychological Assessment*, 6(3), 255–270. <https://doi.org/10.1037/1040-3590.6.3.255>
- Sass, D. A., Schmitt, T. A., & Marsh, H. W. (2014). Evaluating model fit with ordered categorical data within a measurement invariance framework: A comparison of estimators. *Structural Equation Modeling*, 21(2), 167–180. <https://doi.org/10.1080/10705511.2014.882658>
- Satherley, N., Milojev, P., Greaves, L. M., Huang, Y., Osborne, D., Bulbulia, J., & Sibley, C. G. (2015). Demographic and psychological predictors of panel attrition: Evidence from the New Zealand Attitudes and Values Study. *PLOS ONE*, 10(3), Article e0121950. <https://doi.org/10.1371/journal.pone.0121950>
- Sharp, C., Goodyer, I. M., & Croudace, T. J. (2006). The Short Mood and Feelings Questionnaire (SMFQ): A unidimensional item response theory and categorical data factor analysis of self-report ratings from a community sample of 7-through 11-year old children. *Journal of Abnormal Child Psychology*, 34(3), 365–377. <https://doi.org/10.1007/s10802-006-9027-x>
- Shore, L., Toumbourou, J. W., Lewis, A. J., & Kremer, P. (2018). Longitudinal trajectories of child and adolescent depressive symptoms and their predictors—a systematic review and meta-analysis. *Child and Adolescent Mental Health*, 23(2), 107–120. <https://doi.org/10.1111/camh.12220>
- Solmi, M., Radua, J., Olivola, M., Croce, E., Soardo, L., de Pablo, G. S., & Fusar-Poli, P. (2021). Age at onset of mental disorders worldwide: Large-scale meta-analysis of 192 epidemiological studies. *Molecular Psychiatry*, 27, 281–295. <https://doi.org/10.1038/s41380-021-01161-7>
- Stochl, J., Khandaker, G. M., Lewis, G., Perez, J., Goodyer, I. M., Zammit, S., & Jones, P. B. (2015). Mood, anxiety and psychotic phenomena measure a common psychopathological factor. *Psychological Medicine*, 45(7), 1483–1493. <https://doi.org/10.1017/S003329171400261X>
- Stringaris, A. (2011). Irritability in children and adolescents: A challenge for DSM-5. *European Child & Adolescent Psychiatry*, 20(2), 61–66. <https://doi.org/10.1007/s00787-010-0150-4>
- Thabrew, H., Stasiak, K., Bavin, L. M., Frampton, C., & Merry, S. (2018). Validation of the Mood and Feelings Questionnaire (MFQ) and Short Mood and Feelings Questionnaire (SMFQ) in New Zealand help-seeking adolescents. *International Journal of Methods in Psychiatric Research*, 27(3), Article e1610. <https://doi.org/10.1002/mpr.1610>
- Thapar, A., Eyre, O., Patel, V., & Brent, D. (2022). Depression in young people. *The Lancet*, 400, 20–26. [https://doi.org/10.1016/S0140-6736\(22\)01012-1](https://doi.org/10.1016/S0140-6736(22)01012-1)
- Thapar, A., & McGuffin, P. (1998). Validity of the shortened Mood and Feelings Questionnaire in a community sample of children and adolescents: A preliminary research note. *Psychiatry Research*, 81, 259–268. [https://doi.org/10.1016/S0165-1781\(98\)00073-0](https://doi.org/10.1016/S0165-1781(98)00073-0)
- Turner, N., Joinson, C., Peters, T. J., Wiles, N., & Lewis, G. (2014). Validity of the Short Mood and Feelings Questionnaire in late adolescence. *Psychological Assessment*, 26(3), 752–762. <https://doi.org/10.1037/a0036572>
- Vos, T., Barber, R. M., Bell, B., Bertozzi-Villa, A., Biryukov, S., Bolliger, I., & Duan, L. (2015). Global, regional, and national incidence, prevalence, and years lived with disability for 301 acute and chronic diseases and injuries in 188 countries, 1990–2013: A systematic analysis for the Global Burden of Disease Study 2013. *The Lancet*, 386(9995), 743–800. [https://doi.org/10.1016/S0140-6736\(15\)60692-4](https://doi.org/10.1016/S0140-6736(15)60692-4)
- Weyandt, L. L., Iwaszuk, W., Fulton, K., Ollerton, M., Beatty, N., Fouts, H., Schepman, S., & Greenlaw, C. (2003). The internal restlessness scale: Performance of college students with and without ADHD. *Journal of Learning Disabilities*, 36(4), 382–389. <https://doi.org/10.1177/00222194030360040801>
- White, I. R., Royston, P., & Wood, A. M. (2011). Multiple imputation using chained equations: Issues and guidance for practice. *Statistics in Medicine*, 30(4), 377–399. <https://doi.org/10.1002/sim.4067>
- White, J., Zaninotto, P., Walters, K., Kivimäki, M., Demakakos, P., Biddulph, J., & Batty, G. D. (2016). Duration of depressive symptoms and mortality risk: The English longitudinal study of ageing (ELSA). *The British Journal of Psychiatry*, 208(4), 337–342. <https://doi.org/10.1192/bjp.bp.114.155333>
- Widaman, K. F., & Revelle, W. (2022). Thinking thrice about sum scores, and then some more about measurement and analysis. *Behavior Research Methods*. Advance online publication. <https://doi.org/10.3758/s13428-022-01849-w>

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